

2.5.1 Linear Data Structures: Array

ایرے ایک جیسے عناصر کو میموری کے لگاتار بلاک میں رکھتا ہے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 15 of 24

2.5.1 Linear Data Structures: Array

LEARN IT

An **array** holds a collection of **similar items** (the same data type) placed next to each other in **contiguous memory** (one continuous block).

Each item has a position number called an **index**, usually starting from **0**. Because the address of any cell can be calculated from the index, you can jump straight to it. That is **direct access** (also called random access), and it is $O(1)$.

Array name: A

23	76	12	90	34	68
0	1	2	3	4	5

A[3] = 90 (index starts at 0)

Six integers in contiguous cells. Direct access: A[3] jumps to 90 without walking 23, 76 and 12.

WORKED EXAMPLE

Worked example: read by index.

```
A = [23, 76, 12, 90, 34, 68]
print(A[3])
print(A[0])
```

OUTPUT

```
90
23
```

REMEMBER IT

Index 0 is the first shelf on the row. The third item that humans call "number 4" is index 3. Formula in your head: **position from the left, minus one.**

COMMON MISTAKE

Limitation. Arrays are fast to read by index but **inflexible** to grow in the middle. Inserting a new value between 12 and 90 means shifting 90, 34 and 68 one cell to the right. That shift is $O(n)$, so arrays are a poor fit when you insert and delete in the middle all the time.

اردو میں سمجھیں

ایرے ایک جیسے عناصر کو میموری کے لگاتار بلاک میں رکھتا ہے۔ ہر عنصر کا انڈیکس (عموماً 0 سے) ہوتا ہے جس سے براہ راست رسائی میں مل جاتی ہے، جیسے $A[3] = 90$ ۔ درمیان میں نیا عنصر ڈالنے کے لیے دائیں طرف شفٹ کرنا پڑتا ہے جو سست ہے۔ $O(1)$

CHECK YOURSELF

1. What does "contiguous memory" mean?
2. In the diagram, what is A[2]?
3. Why is inserting in the middle of an array inefficient?

Answers: 1. The items sit in one continuous block, next to each other. 2. 12. 3. Later elements must all be shifted right to make room.

Next lecture: Linear Data Structures: Linked List